



Current status of *Acropora palmata* and *Acropora cervicornis* in the Colombian Caribbean: demography, coral cover and condition assessment

Rocío García-Urueña · Marco A. Garzón-Machado

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Abstract *Acropora* provides an example of reef degradation, as evidenced by population reductions in Caribbean reefs. However, data on the current status of *Acropora* populations in Colombia are lacking. In this study, reef habitats were surveyed for *Acropora palmata* and *Acropora cervicornis* throughout the Colombian Caribbean, and the size structure, coverage and colony conditions of their populations were evaluated. *Acropora palmata* size classes were negatively skewed, indicating low recruitment. This species exhibited a patch-type distribution around Tayrona, Isla Fuerte and Isla Arena; these localities exhibited the highest cover among the surveyed regions. The lowest coral cover was recorded for the Rosario Islands. Healthy colonies were dominant; however, disease and damselfish territories were common. The size class data of *Acropora cervicornis* indicated the persistence of small colonies; however, no recruits were observed. The distribution was dispersed, with important patches and the highest cover observed around the Urabá Gulf and Rosario

Islands. Overall, the cover was lower than 10%. Few healthy colonies were observed, and macroalgae and sponges were common. Conservation and management efforts are required for both species; for *Acropora palmata*, efforts targeting the Rosario Islands are needed, whereas *Acropora cervicornis* is in critical condition throughout the Colombian Caribbean.

Keywords Endangered species · Elkhorn coral · Population dynamics

Introduction

Acropora palmata (Lamarck, 1816) and *A. cervicornis* (Lamarck, 1816) are two species distributed in the western Atlantic and the Caribbean, and their deterioration and population declines have been an important topic of discussion in reef conservation. The formal inclusion of these species in the IUCN Red List of Threatened Species (Aronson et al., 2008a, b) has increased concern over and focused attention on the general decline of these coral populations. In the reef areas of the Colombian Caribbean, the average live coral cover ranges from 9 to 43% (Rodríguez-Ramírez et al., 2006). In Tayrona National Natural Park (TNNP), Garzón-Ferreira and Cano (1991) reported 60% mortality of *A. palmata* and 80% mortality of *A. cervicornis*, due to increased sedimentation, reduced light penetration, nutrient enrichment, disease and

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R. García-Urueña (✉) · M. A. Garzón-Machado
University of Magdalena, Santa Marta 470004, Colombia
e-mail: rgarciau@unimagdalena.edu.co

dynamite fishing. Both species have reached mortalities of 90% in Corales del Rosario National Natural Park (Díaz et al., 2000), possibly due to the seasonal influences of turbid, nutrient-rich and low-salinity waters from Cartagena Bay and the Dique Channel (Alvarado et al., 1986). According to Ardila et al. (2002), in Colombia, these two species are classified in the red book as endangered and critically endangered. The deterioration has been attributed to factors such as climate change, coastal development, overfishing and the impacts of mass tourism practices (Buddemeier et al., 2004). Massive bleaching events (Hoegh-Guldberg, 1999) and the appearance of specific diseases in acroporids, such as white band disease (WBD) (Porter et al., 2001) and white pox (WPX) (Patterson et al., 2002), have also intensified coral mortality (Muller et al., 2008; Rogers & Muller, 2012). Despite the high growth rates of these species, the last assessments in Colombia (Díaz et al., 2000; Garzón-Ferreira et al., 2004) showed no evidence of recovery.

Coral reef areas of the Colombian Caribbean occupy an area of 2860 km² distributed in 26 discrete areas; among them, the oceanic reefs in the archipelago of San Andrés and Providencia are the most prominent and complex reef ecosystems. These areas are present along the coasts and scattered on the continental shelf from the Urabá Gulf, which is located very close to the border of Panama, to the northern region of the Guajira Peninsula (Díaz et al., 2000). Although the degradation problem is similar among coral reef areas, the sedimentary discharge varies, which influences the deterioration of populations. The Urabá Gulf receives sediments from the Atrato River, and Isla Fuerte, which is located approximately 11 km off the coast, is a small platform influenced by the Sinú River. The Sinú and Magdalena Rivers influence the San Bernardo and Rosario Islands, respectively, specifically near the Canal del Dique in the Rosario Islands. Coral areas such as Isla Arena are influenced by the sediment loads of the Magdalena River, whose mouth is located 40 km to the NE. In the area of Santa Marta and TNNP, the trade winds lead to upwelling, which causes decreases in temperature to as low as 20°C (Bayraktarov et al., 2014).

Acropora populations in Colombia have been evaluated in Isla Arena (Pinzón et al., 1998) and TNNP (Garzón-Ferreira et al., 2004), and monitoring

programs have provided insights into changes in their coverage (National coral reef monitoring system in Colombia, SiSMAC). Unfortunately, the assessments were performed at specific locations to monitor cover changes, and although the data showed declining population trends, extensive areas of the reefs have yet to be explored. Assessments of the current status of *Acropora* species have been conducted in Florida, Puerto Rico, the Virgin Islands (Rogers & Muller, 2012; Wirt et al., 2013, 2015), Mesoamerican reefs (Rodríguez-Martínez et al., 2014) and Los Roques in Venezuela (Zubillaga et al., 2008; Croquer et al., 2016); however, in Colombia, no recent evaluations have been made.

To implement effective conservation and management programs for reef areas, the current distribution of *Acropora* must be determined. As life history and coral population dynamics are strongly related to size-frequency distributions, these variables are widely used as a simple and non-destructive method to study the population structure of *Acropora* species; such analyses may reveal the long-term consequences of the heterogeneous structure of populations under different environmental influences (Bak & Meesters, 1998; Meesters et al., 2001; Vermeij & Bak, 2003). Similarly, a demographic analysis can reveal the extent to which these organisms are under stress and is especially useful when anthropogenic activities affect the reefs and the viability of coral populations (Smith et al., 2005).

The size structures, coverage and conditions of the coral populations were assessed in seven localities along a latitudinal gradient that differed in their degree of anthropogenic disturbance in the Colombian continental Caribbean and the insular area of San Andrés. For this purpose, we evaluated previous reports and performed field verifications to determine the distributions of potential *A. palmata* and *A. cervicornis* habitats. We aimed to answer the following question: How do remnant populations of *Acropora* species vary across sites in the Colombian Caribbean in size-frequency distribution, cover and condition? The findings reported herein provide insight into the status of *A. palmata* and *A. cervicornis* in the Colombian Caribbean. This study provides baseline data for future comparisons and can help guide local management and conservation efforts.

Materials and methods

Study areas

Seven localities that are potential habitats for *Acropora* species situated on an insular or continental shelf along the Colombian Caribbean were studied. 1. The TNNP (11°15'20"–11°22' N and 73°57'–74°12' W) is located on the eastern coast and harbors two types of coral formations: hermatypic corals that have partially colonized slabs of rock fallen from the cliffs and can be found between 5 and 8 m deep, and more developed and compact formations built on more stable rocky substrate and in areas less exposed to the waves (Díaz et al., 2000). 2. Isla Arena (10°44'20" N and 75°21'10" W) is a small emergent shoal formed by the accumulation of litho-bioclastic sediments and calcareous debris; it is located on the continental shelf approximately 8 km from the coast. 3. The Rosario Islands (10°07'–10°14' N and 75°37'–75°57' W) present coral formations in an advanced state of degradation. Living formations are found along the NE flank of the Barú Peninsula and form several banks on the high relief of the continental shelf (Díaz et al., 2000). 4. The archipelago of San Bernardo (9°39'–9°52' N and 74°40'–74°57' W) is composed of nine islands of coral origin and an artificial islet built by fishermen with conch shells and coral debris (Díaz et al., 2000). The coral formations with greater development and coverage are located on the north and west flanks, where fringing reefs and coral patches up to 30 m deep predominate. 5. Isla Fuerte is situated in the central region (9°23'N and 76°11'W) 11 km from the coast; the island has a seaward fringing barrier reef, a peripheral reef-belt at the outer shelf margin, and shallow patchreefs (Díaz et al., 1996). 6. The Urabá Gulf (8°35'–8°41' N and 77°15'–77°22' W) is approximately 85 km in length and 15–30 km in width along the northeastern margin near the border with Panama and presents patch-like or fringing formations (Díaz et al., 2000). 7. San Andrés Island (SAI) (12°28'–12°37' N and 81°39'–81°45' W) has coral reefs over an area of 44.7 km², including an oceanic reef complex with a barrier reef, an insular platform reef, fringe reefs and lagoon patches (Díaz et al., 2000) (Fig. 1).

Field surveys

Before the field work, databases from the Atlas of Coral Reefs of Colombia (Díaz et al., 2000), articles, technical reports and unpublished data were consulted to identify the locations of the patches or reef sites in the study area (Pinzón et al., 1998; Garzón-Ferreira et al., 2004), and reference maps were then made. Field surveys were conducted between October 2016 and August 2018, and exploration dives were carried out on each reef. Most of the consulted literature provided coral cover information from monitoring programs, but living colonies were generally not found in these locations. Therefore, local fishermen were consulted to find patches and isolated colonies of both *Acropora* species.

In each locality where living colonies of *Acropora* species were found, coordinates were recorded using a Garmin eTrex 20 GPS, and the assessments were performed. For *Acropora palmata*, we surveyed 53 shallow reefs (26 in TNNP, 1 in Isla Arena, 3 in the Rosario Islands, 6 in the San Bernardo Islands, 5 in Isla Fuerte, 8 in the Urabá Gulf and 4 in SAI). For *A. cervicornis*, 24 reefs were surveyed (3 in TNNP, 3 in the Rosario Islands, 6 in the San Bernardo Islands, 5 in Isla Fuerte, 6 in the Urabá Gulf and 1 in SAI). After locating the isolated colonies or patches of *Acropora* species, the percentage cover was quantified. Depending on the size of the patch, between one and five linear transects that were 10 m in length (between 2 and 10 m deep) were randomly deployed over the patch, and videos were recorded.

In addition, photographs and videos of the coral populations were taken to assess the condition according to the following: healthy, presence of *Coralliophila erosa* (Röding, 1798) or *Hermodice carunculata* (Pallas, 1766), sponges, cyanobacteria, macroalgae, disease (WPX or WBD), bleaching, territories of *Stegastes* sp. (Jenyns, 1840) (defined by chimney structures on live coral tissue) and predation by *Acanthostracion* sp. (Bleeker, 1865) (evidenced as white halo rings).

Data analysis

To determine the size structure, we used the maximum diameter (L) and the perpendicular diameter (I) to calculate the area of each colony with the assumption of an ellipse $A = (L/2 \times I/2) \times \pi$ (Yap et al., 1992;

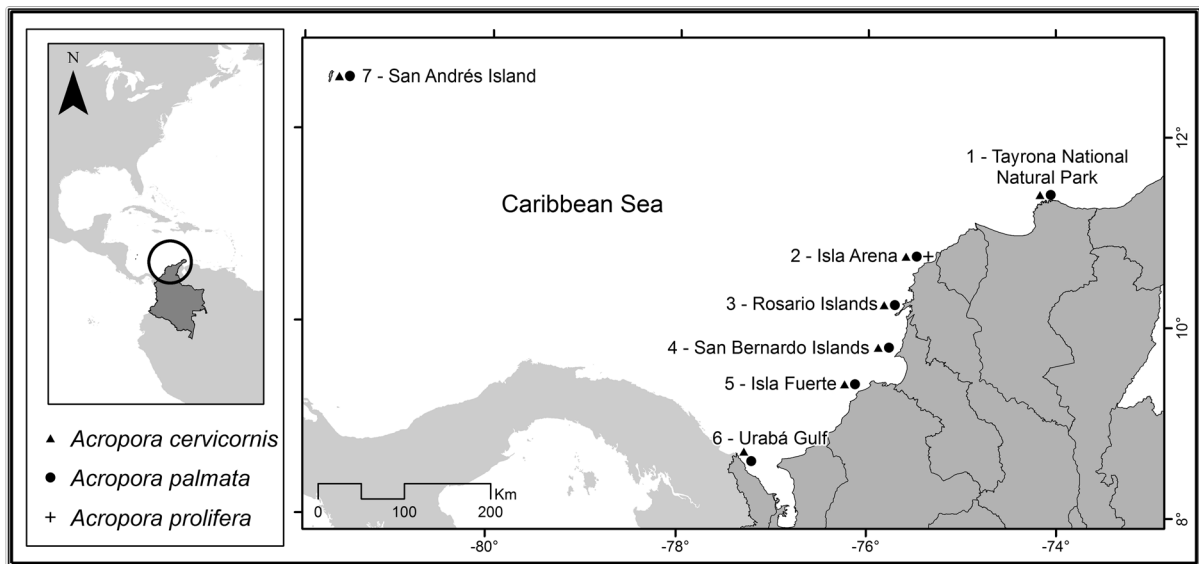


Fig. 1 Study localities in the Colombian Caribbean

Linares et al., 2011). When large patches were found, measurements of the length and width of the patch were taken. Differences in colony size (cm^2) among localities were evaluated by Kruskal–Wallis test (the data presented homogeneity of variance, but were not normally distributed according to the Shapiro–Wilk test at $P < 0.05$ even transformed) followed by Dunn’s post-hoc test. The statistical analyses were performed using SPSS.

The size structures of the populations of *A. palmata* and *A. cervicornis* were analyzed for each of the localities by determining size-frequency distributions and calculating descriptive statistics of mean size, standard deviation, coefficient of variation (CV) and minimum and maximum sizes. The skewness (g_1) was used to quantify the relationship between the numbers of small and large colonies within each population, and kurtosis (g_2) was used to describe the data according to a normal distribution (Anderson & Practchett, 2014). The colony surface area data were $\log_{10}$ transformed to achieve a normal distribution and to increase the resolution of small sizes (Bak & Meesters, 1998; Vermeij & Bak, 2003). The Kolmogorov–Smirnov test using Lilliefors correction was used to test the normality of the size-frequency distribution of each species. Similarities between localities were calculated with the Spearman rank-correlation coefficient (12 size classes based on a

logarithmic scale in *A. palmata* and 10 in *A. cervicornis*).

To calculate the coverage, videos from each reef were analyzed using the Coral Point Count program. Twenty images were captured from each video transects with 5-s intervals of the total video time; and the coverages of the substrate components were then estimated (Kohler & Gill, 2006). The results are presented as the mean $\pm$ SD. The assumptions of normality and homoscedasticity were tested and passed using the Shapiro–Wilk test and Levene test, respectively. For *A. palmata*, was necessary the arcsine transformed. Differences among localities for each species were analyzed by ANOVA.

Data for the condition assessment of each species were summarized with descriptive statistics, and the percentage of each condition in each locality was calculated and plotted in pie charts.

Results

Acropora palmata

For *A. palmata*, 1292 colonies were measured. The untransformed data showed that small colonies ($28\text{--}141\text{ cm}^2$) were found at all sampling sites, while large colonies of more than $10,000\text{ cm}^2$ were found only at the Isla Fuerte and TNNP sites. The Kruskal–

Wallis test of colony size showed significant differences among locations. Eleven of the 21 comparisons performed by Dunn's test showed significant differences.

The log-transformed data showed 12 size classes, ranging between 1.45 and 5.71 cm², and were normally distributed in only the Urabá Gulf, Isla Arena and SAI (Table 1). Colonies of the largest size were at the Isla Fuerte and TNNP (16.7 and 7.2%, respectively), and all localities (except Isla Arena) presented a low percentage (between 1.2 and 3%) of small size classes (classes 1 and 2; < 200 cm²). The Spearman rank-correlation coefficient analysis of size-frequency distributions showed that the Isla Fuerte site was significantly different from the other sites except the TNNP and Isla Arena (Fig. 2). The CV was similar among all locations, ranging from 14.3% for Isla Arena to 18.7% for TNNP. In addition, a negative skewness (g_1) was dominant, with values between -0.73 in the San Bernardo Islands and -0.23 in TNNP, indicating the predominance of large colonies in all localities (Table 1).

The most important patch-type formations of *A. palmata* were located in TNNP, and small isolated patches were observed in Isla Arena, the Urabá Gulf, Isla Fuerte and the San Bernardo Islands. The rest of the study area was mainly dominated by scattered colonies (Supplementary Material S1). The Rosario Islands, where the main formation was previously located on the crest of the Isla Grande reef, is currently made up of dead colonies, with only 4.94% coverage of *A. palmata*. The highest coverage was found at the TNNP and Isla Arena sites (21.86% and 24.25%, respectively) (Fig. 3); however, cover has increased at TNNP from 10% in 2004, whereas that at Isla Arena has decreased from 80% in 1998. No differences were found in *A. palmata* cover among the localities (ANOVA, $F_{5,38} = 1.54$, $P > 0.020$).

Healthy colonies dominated in all localities, with values fluctuating between 79% (Isla Arena) and 31% (Rosario Islands). *Stegastes* sp. territories and *Acanthostracion* sp. predation were the second most common conditions, occurring in 7% to 34% of colonies. Diseased colonies represented an important condition in TNNP and the Rosario Islands at 26%,

Table 1 Summary statistics of colony size-frequency distributions for *Acropora palmata* and *A. cervicornis* at each locality (the Urabá Gulf (UG), Isla Arena (IA), Isla Fuerte (IF),

the Rosario Islands (RI), San Bernardo Islands (SBI), Tayrona National Natural Park (TNNP) and San Andrés Island (SAI))

Species	Location	<i>n</i>	Mean colony size m ²	SD	CV	Min	Max	g_1	g_2	P_{norm}
<i>Acropora palmata</i>	TNNP	545	3.67	0.69	19	1.64	5.46	-0.232	0.199	0.000
	IA	45	3.92	0.56	14	2.58	4.86	-0.542	-0.494	0.200
	IR	101	3.50	0.59	17	1.64	4.54	-0.559	0.306	0.047
	ISB	167	3.53	0.52	15	1.80	4.59	-0.734	1.245	0.012
	IF	84	3.98	0.66	17	2.07	5.32	-0.681	0.557	0.034
	GU	278	3.45	0.59	17	1.45	4.75	-0.430	0.22	0.051
	SAI	72	3.31	0.55	17	2.05	4.39	-0.243	-0.403	0.200
	Total	1292	3.59	0.64	18	1.45	5.46	-0.247	0.157	0.000
<i>Acropora cervicornis</i>	TNNP	48	2.27	0.38	17	1.36	2.95	-0.351	-0.525	0.440
	IR	82	2.33	0.43	18	1.54	3.47	0.397	-0.207	0.253
	ISB	54	2.00	0.28	14	1.30	2.49	0.000	-0.579	0.341
	IF	49	1.83	0.34	19	1.18	2.65	0.537	0.075	0.115
	GU	45	2.41	0.50	21	1.34	3.46	0.243	-0.368	0.440
	SAI	13	2.56	0.39	15	1.84	3.22	-0.188	-0.408	0.997
	Total	291	2.18	0.45	20	1.18	3.47	0.431	-0.131	0.006

The sample size (*n*), log-transformed mean colony size, standard deviation (SD), coefficient of variation (CV), minimum and maximum colony size (Min, Max), skewness (g_1) and kurtosis (g_2) are shown. p_{norm} : probability that data are normally distributed (Kolmogorov–Smirnov test with Lilliefors adjustment for *A. palmata* and Shapiro–Wilk for *A. cervicornis*)

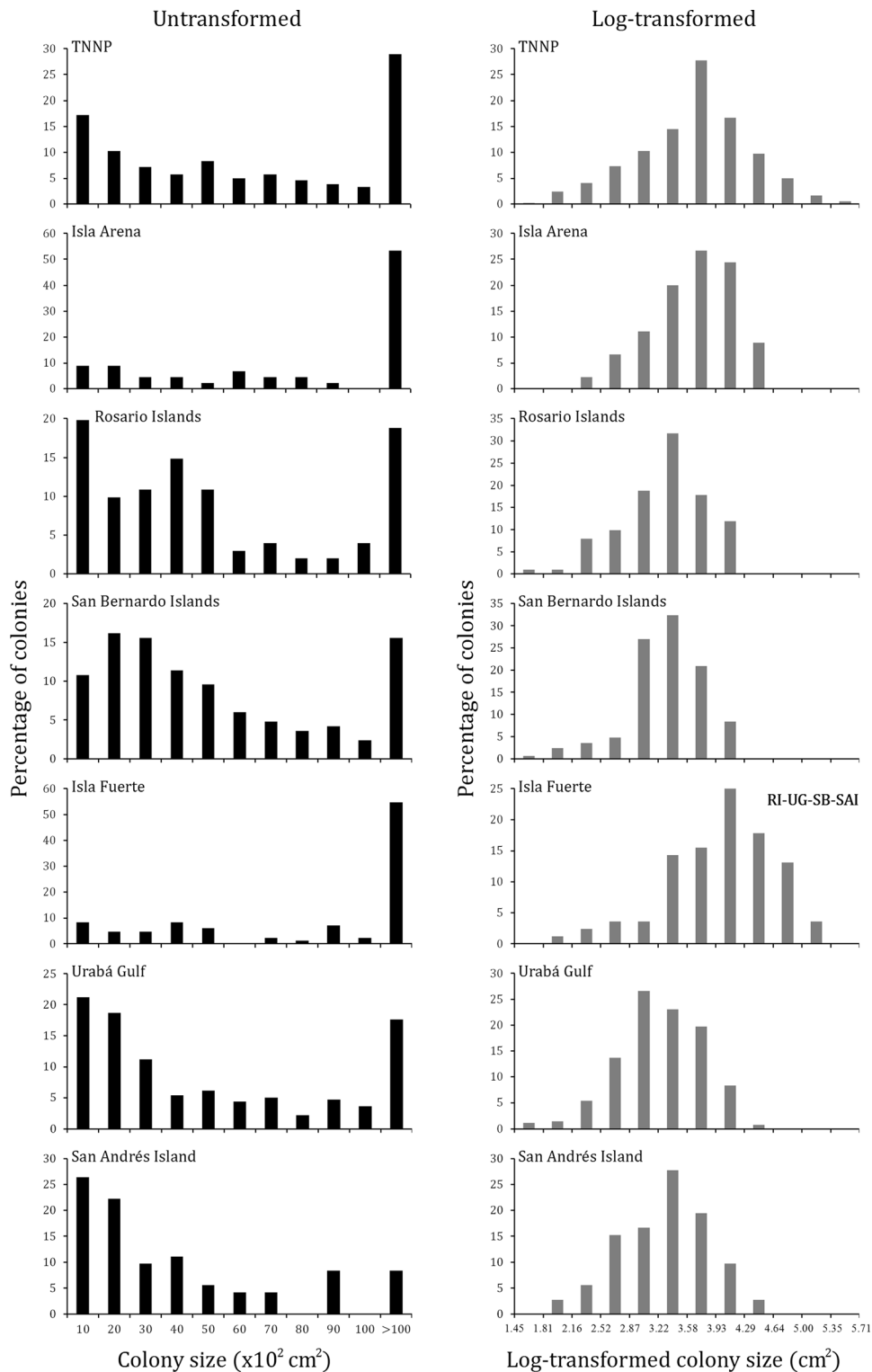


Fig. 2 Untransformed and log-transformed size-frequency distributions of *Acropora palmata*. Localities with significantly different distributions (Kolmogorov–Smirnov tests, $P > 0.05$) are indicated on each graph

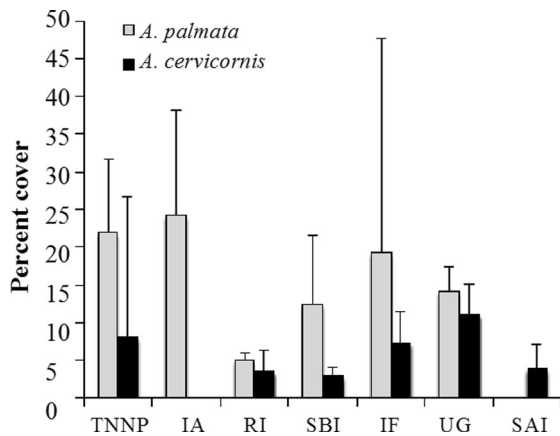


Fig. 3 Percent cover of *Acropora palmata* and *A. cervicornis* in the studied locations ($\pm$ SD) of Tayrona National Natural Park (TNNP), Isla Arena, the Rosario Islands (RI), San Bernardo Islands (SB), Isla Fuerte (IF), the Urabá Gulf (UG), and San Andrés Island (SAI)

followed by Isla Fuerte and the San Bernardo Islands at 15% and 14%, respectively. Bleaching was present in only Isla Arena (30%), the Urabá Gulf (< 1%) and San Bernardo Islands (< 1%). The presence of macroalgae was important in SAI (14%) and the San Bernardo Islands (13%); sponges were present in the San Bernardo Islands, the Rosario Islands, Isla Fuerte and the Urabá Gulf (12, 4, 7 and 8%, respectively), and *C. erosa* was observed in only SAI (14%) (Fig. 4).

Acropora cervicornis

For *A. cervicornis*, 291 colonies were measured; generally, small or relict colonies (< 30 cm²) predominated in all localities. Small patches in good condition were present only at Isla Rosario reef in the Rosario Islands and in the Palmar and La Mora reefs in the Urabá Gulf. The Kruskal–Wallis test showed significant differences among locations in colony size. Eight of 15 comparisons performed by Dunn's test showed significant differences, with Isla Fuerte and San Bernardo Islands showing differences from the rest of the localities but not between each other. The transformed data varied between 1.18 and 3.66 cm² (untransformed range: 15–3000 cm²) and were grouped into 10 intervals (Fig. 5). The distributions were normally distributed at all localities (Table 1). The Spearman rank-correlation analysis revealed that Isla Fuerte significantly differed from all of the remaining localities, except the San Bernardo Islands;

the size-frequency distributions were similar between these two locations. In addition, the distribution at the San Bernardo Islands was significantly different from the Urabá Gulf (Fig. 5). A positive skewness tendency, indicating a predominance of small colonies, was observed for all localities except TNNP and SAI, which exhibited negative skew; however, the larger colonies did not exceed 160 cm². The lowest CV corresponded to the San Bernardo Islands (14%), whereas the highest corresponded to the Urabá Gulf (20.8%) (Table 1).

The percentage coverage of *A. cervicornis* was low in all locations. The highest coverage was observed in the Urabá Gulf (11%); Isla Fuerte and TNNP presented values of 7.25% and 7.96%, respectively. The Rosario Islands, San Bernardo Islands and SAI presented values below 4%. Isla Arena was the locality with the most extreme change in coverage, which decreased from 90% in 1998 to 0.0% in this study. In the Rosario Islands, the coverage of coral species was 30.49%; coverage ranged between 14 and 24% at the remaining localities (Fig. 3). No differences were found among localities (ANOVA, $F_{5,10} = 1.78$, $P = 0.204$).

More types of conditions, mainly involving the presence of macroalgae, sponges and invertebrates, were observed for *A. cervicornis* than for *A. palmata*. Diseases were not observed, and bleaching occurred in only TNNP (6%), Isla Fuerte (< 1%) and the Urabá Gulf (< 1%). More than 50% of the populations in the SAI, Urabá Gulf and the Rosario Islands were classified as healthy, whereas in TNNP, the San Bernardo Islands and Isla Fuerte, showed values between 28 and 38%, respectively. The presence of macroalgae was important, with a high proportion of colonies being observed under this condition (between 19 and 40% in all localities). The presence of colonies with sponges was important in San Bernardo Islands (43%), Isla Fuerte (15%) and the Urabá Gulf (11%). *C. erosa* exhibited high values in TNNP (22%), the Urabá Gulf (11%) and Isla Fuerte (6%). *Stegastes* sp. territories exhibited the highest presence in TNNP (14%), whereas in the remaining localities, between 4 and 5% of colonies exhibited *Stegastes* sp. presence. No signs of *Acanthostracion* sp. were evident (Fig. 4).

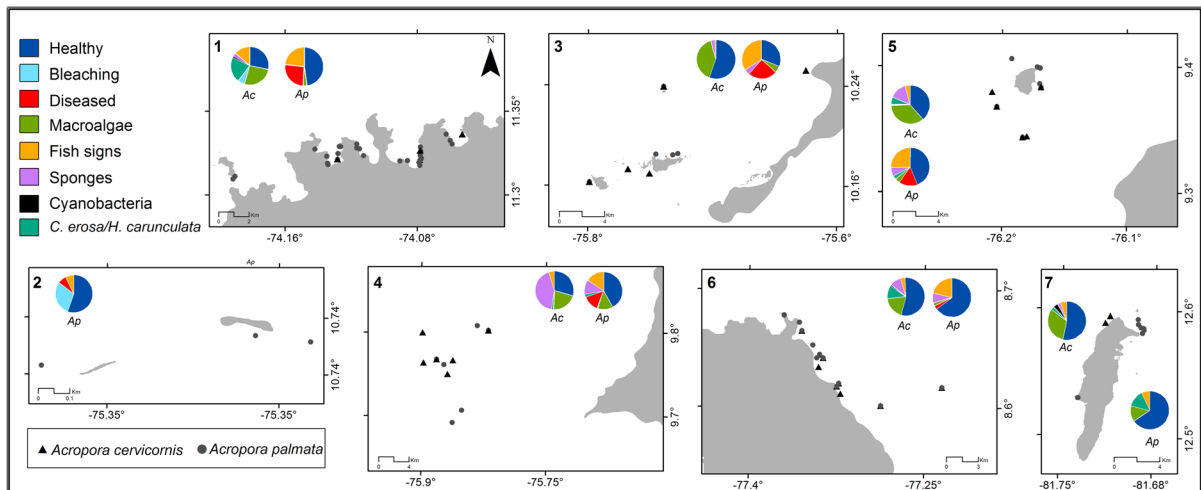


Fig. 4 Percentages of colony conditions surveyed for *Acropora palmata* and *A. cervicornis* in the Colombian Caribbean. 1. Tayrona National Natural Park (TNNP), 2. Isla Arena, 3. The

Rosario Islands (RI), 4. San Bernardo Islands (SB), 5. Isla Fuerte (IF), 6. the Urabá Gulf (UG), and 7. San Andrés Island (SAI)

Discussion

This work aimed to establish the current status of *A. palmata* and *A. cervicornis* populations throughout the Colombian Caribbean. Patch-type formations were important for *A. palmata* in TNNP and Isla Arena, and they were found scattered along the rest of the study area. However, the presence of small-sized colonies suggests signs of recovery in some localities as well as possible stabilization in cover. For *A. cervicornis*, the situation is critical, and a reduction of 90% was observed in TNNP, where only one formation remains from those that were studied by Garzón-Ferreira et al. (2004). Likewise, this species has nearly disappeared in Isla Arena. In the remainder of the Colombian Caribbean, its coverage does not exceed 11%. The largest patch is located in Cayo Bolívar, north of SAI (not assessed in this study), and small populations are present in the Rosario Islands and in the Urabá Gulf.

Transformed size-frequency distribution data showed a negative skewness for *A. palmata*, indicating the predominance of large colonies, while the positive skewness for *A. cervicornis* indicated that small sizes were predominant. The contrasting distributions between species can indicate changes in life history (e.g., growth, partial mortality, recruitment, etc.) and may vary at small spatial scales (Babcock, 1991; Baird & Hughes, 1997; Hughes & Tanner, 2000; Adjeroud et al., 2007). Meesters et al. (2001) indicated that

environmental degradation can affect population structure. Reefs, whether reefs located near populated areas may exhibit few colonies of small-size classes and numerous large colonies, represent negative skewness. In our study, the differences between *Acropora* species indicate species- and site-specific responses; marked contrasts were observed in TNNP, which showed larger sizes for *A. palmata* but a near 90% decline for *A. cervicornis*, and the Rosario Islands, with important patches of *A. cervicornis* but a near total absence of *A. palmata*. Differences in local effects, such as the discharges of major rivers, distances to these sources of discharge, changes in temperature, and frequencies of these disturbances, affect the colonies on a small scale and determine general changes in observed patterns (Hughes, 1984b; Wallace, 1985). However, in our study, it was difficult to determine whether partial mortality or any other factor reduced the colony sizes. Further, the differences cannot be directly attributed to the latitude gradient or to responses to environmental changes: some locations, such as the Urabá Gulf, showed similarities in size-frequency distribution to both distant locations (≈ 450 km), such as TNNP, and closer locations, such as Isla Fuerte (≈ 140 km). Therefore, our study provides a starting point to evaluate the life histories of these coral species.

In recent decades, coral reef coverage has been evaluated in insular and continental reef areas of the

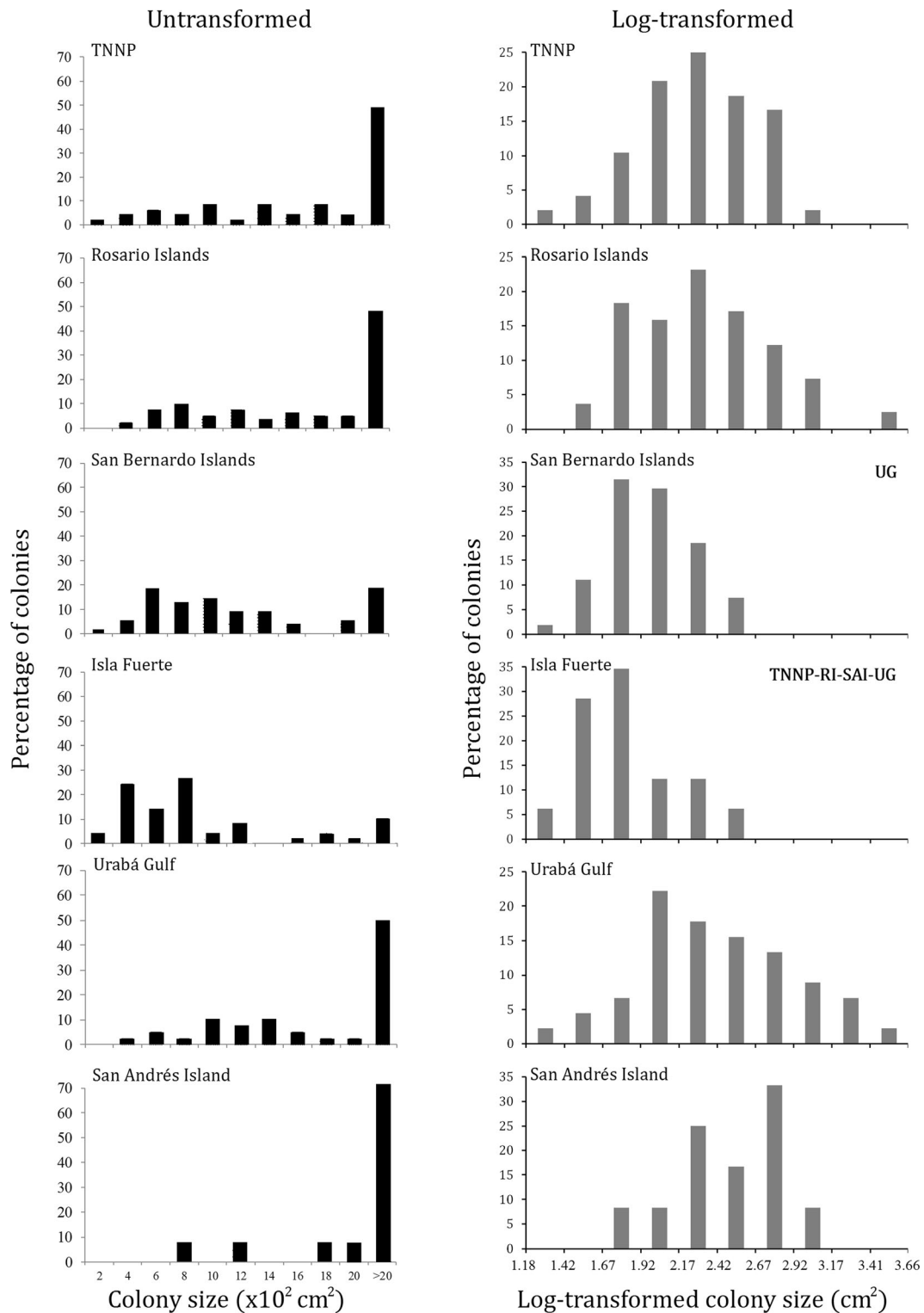


Fig. 5 Untransformed and log-transformed size-frequency distributions of *Acropora cervicornis*. Localities with significantly different distributions (Kolmogorov–Smirnov tests, $P > 0.05$) are indicated on each graph

Colombian Caribbean (Martínez & Acosta, 2005; Rodríguez-Ramírez et al., 2006; Vega-Sequeda et al., 2008), showing significant reductions in comparison to those reported historically. However, specifically for shallow reef areas of acroporid formations, data have been published only for Isla Arena by Pinzón et al. (1998), indicating an average live cover of 76% for *A. palmata* and 14% for *A. cervicornis*. In the Rosario Islands, Cendales et al. (2002) observed acroporid mortality of up to 100%, which was associated with high rates of sedimentation and nutrient input due to the Dique Channel and the colonization of coral debris by macroalgae and other corals, such as *Agaricia tenuifolia*, *A. agaricites*, *Porites astreoides* and *Millepora* spp. Garzón-Ferreira et al. (2004) reported an average live coverage of *A. palmata* of 10% and values between 0.0 and 58.3% in TNNP. For *A. cervicornis*, live coverage reached only 5%, with values of zero in most formations and a maximum of 35.3%.

Acropora palmata populations in the Colombian Caribbean show high abundances of adults and new recruits distributed along the reefs. In specific areas such as TNNP, our results suggest that *A. palmata* cover has increased compared with reported over the last two decades. In fact, the data obtained in our study were compared with historical data collected by Garzón-Ferreira et al. (2004), suggesting that some formations present higher cover. However, in Isla Arena, cover has decreased from 76% (Pinzón et al. 1998) to 17.7% (this study), and the species has almost disappeared from Isla Grande reef in the Rosario Islands. There was no major difference in the percentages of *A. palmata* percentage cover among the remaining locations, which ranged from 12 to 24%. The variation suggests that the effects of small-scale disturbances, such as depredation, fragmentation, sediment accumulation and competitive interactions, need to be assessed to determine whether they affect population coverage. For *A. palmata* in the Rosario Islands, continuous sediment discharges through the Dique channel reduced coral cover (Alvarado et al., 1986). In the remaining localities, although the coverage of this species did not exceed 24%, there are no data on the possible factors that are causing reductions or favoring increases of these populations.

The situation for *A. cervicornis* is critical, as coverage reduction is widespread, but temporal

changes and the gradual degradation rate could not be determined here. The percentages of cover in the 24 reefs evaluated were low, with an overall mean value of 5.6%, and only three localities (San Bernardo Islands, the Rosario Islands and the Urabá Gulf) exhibited important isolated patches, suggesting rapid population declines. In TNNP, Garzón-Ferreira et al. (2004) reported 12 formations with a live coverage of only 5% in certain locations, with values of zero in most formations and a maximum value of 35.3%. In our study, out of these 12 formations, only one was found on the Neguanje reef, with a coverage of 4%. However, two formations were found on the reefs of Chengue and Cinto, which increased the coverage for TNNP to 7.9%. The most extreme case is Isla Arena, where Pinzón et al. (1998) reported 14% coverage and we recorded 0%; however, we did observe two isolated live colonies.

In general, studies on the interactions between *Acropora* spp. and biotic stressors have used measures such as cover in specific areas or monitoring programs, with few studies relating the conditions and health of colonies. Although our results did not show differences among localities, the two species shared almost the same health conditions; however, their presence status and frequencies were different, probably associated with variation in the site at which the coral develops and the tolerance levels of each species (Hall et al., 2012; Nepote et al., 2016). For instance, *A. palmata* showed an important presence of healthy colonies in all localities (50%); this trend may indicate successful recovery in some areas despite the generalized reef degradation in the wider Caribbean (Bud-demeier et al., 2003; Gardner et al., 2003). Nevertheless, certain competitive interactions may contribute to their decline, such as predation by *Hermodice carunculata* and *Coralliophila erosa* and the presence of damselfish territories, *Acanthostracion* sp., sponges and macroalgae, which enhance potential habitat losses for future recruitment (Bruckner, 2002). Bleaching in Isla Arena may be attributed to prolonged calm waters and to high levels of solar irradiance that occur in only November, which is the calmest period of the year to survey the area (Gleason & Wellington, 1993; Banaszak & Lesser, 2009). The observation of WPX and WBD confirms the wide distributions of these epizootic events, which enhance colony susceptibility and thus the reduction in abundance throughout Colombian reefs (Garzón-Ferreira et al., 2001;

Rodríguez-Ramírez et al., 2010). Given the high percentages of damselfish territories in all locations, it is of interest to evaluate the possible consequences at the population level, which may indicate whether the presence of these territories can contribute to colony deterioration.

Acropora cervicornis presented greater variability in the frequency of observed conditions than *A. palmata* with respect to the localities, with colonies principally being affected by algal overgrowth; this supports the hypothesis that the trade-off between algae and coral is a common trend in degraded reefs (McManus & Polsenberg, 2004; Meltvedt & Jadot, 2014). San Bernardo Island also exhibited a notable presence of encrusting and tube-like sponges, possibly due to anthropogenic stressors in the surrounding areas, such as the direct discharge of domestic waste and overpopulation in the surrounding islets. In TNNP, damselfish territories, macroalgae overgrowth and signs of predation by fireworms and mollusks were important (52%); however, damselfish territories were less evident in the other localities, exhibiting frequencies between 1 and 5%. Garzón-Ferreira et al. (2004) reported 79% macroalgae cover in TNNP, while some of the dominant conditions observed in the past (i.e., *Stegastes* and *C. erosa*) are now less important, possibly due to algal overgrowth. Our results show that it is necessary to quantify biological interactions and their possible impact at the population level.

TNNP has outstanding ecological and conservation value for populations of *A. palmata*, having the highest size-class distribution and the most important patch formations among all the localities surveyed. However, more detailed studies are needed to understand the ecological, biological, physiological and environmental conditions that have favored the persistence of *A. palmata* at this locality. For *A. cervicornis*, in areas such as the San Bernardo Islands, specific protection measures are required, such as demarcation of the most important patches as well as regulation of fishing on the reefs. However, each locality may require specific protection measures.

Walker & Gilliam (2013) noted that climate change affects many tropical species, including corals. In Colombia, studies of the coverage and demographics of these species are relevant because they provide baseline data for future comparisons and for developing specific local management and conservation

efforts. While our results define the current status of these species, future studies can refer to our data to determine whether the populations have remained stable and whether changes in competitive interactions and disease occurrence have affected the populations.

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